

Probability HWS

Problem #1

We know Defects per 400 sq ft follows distribution $X \sim \text{Pois}(4.8)$

$$P(X < 6) = P(X \leq 5)$$

$$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}$$

$$P(X \leq 5) = \sum_{n=0}^5 e^{-4.8} \frac{4.8^n}{n!} = e^{-4.8} \left(1 + 4.8 + \frac{4.8^2}{2!} + \frac{4.8^3}{3!} + \frac{4.8^4}{4!} + \frac{4.8^5}{5!} \right) \\ \approx 0.6510$$

Probability that at least two out of six have fewer than 6blemish
success = $P(X \leq 5) = 0.6510$

This follows binomial distribution: $Y \sim \text{Binomial}(n=6, p=0.6510)$

$$P(Y \geq 2) = 1 - P(Y < 2) = 1 - [P(Y=0) + P(Y=1)]$$

$$P(Y=0) = (1-p)^6 = (0.349)^6$$

$$P(Y=1) = 6p(1-p)^5 = 6 \cdot 0.651 \cdot (0.349)^5$$

$$1 - P(Y=0) - P(Y=1) \approx 0.978$$

Problem #2

a) A = a couple is both born on April 30

$$P(A) = \frac{1}{365} \cdot \frac{1}{365} = \frac{1}{133225}$$

for probability $P(\geq 1) = 1 - (1-p)^N$ for $N=80,000$

since $p = \frac{1}{133225}$ and $N=80,000$, it is better to use Poisson distribution.

with $(1-p)^N = e^{-\lambda}$ $\lambda = Np = 80,000 \cdot \frac{1}{133225} = 0.6005$

$$P(\geq 1) = 1 - e^{-\lambda} = 1 - e^{-0.6005} \approx 0.45$$

b) We assume that we can ignore leap years, birthdays are uniform over 365 days

$$P(\text{both partners born on same day}) = \frac{1}{365}$$

$$\lambda = Np = 80,000 \cdot \frac{1}{365} = 219.178$$

$$P(\geq 1) = 1 - e^{-\lambda} = 1 - e^{-219.178} \approx 1.000$$

Problem #3

$X \sim \text{Geo}(p)$ for $p \in (0, 1)$ $P(X=k) = p(1-p)^{k-1}$

$$P(X > k) = \sum_{j=k+1}^{\infty} p(1-p)^{j-1} = (1-p)^k$$

$$P(X > m+n | X > m) = \frac{P(X > m+n)}{P(X > m)} = \frac{(1-p)^{m+n}}{(1-p)^m} = (1-p)^n = P(X > n)$$

$\therefore$ the geometric distribution X is memoryless

Problem #4

Let $X = \#$ of transistors that work in the sample

$X \sim \text{HGeom}(N=100, K=80, n=10)$

$$P(X=k) = \frac{\binom{80}{k} \binom{20}{10-k}}{\binom{100}{10}} \quad k = 0, 1, 2, \dots, 10$$

$$P(X \geq 9) = P(X=9) + P(X=10) = \frac{\binom{80}{9} \binom{20}{1}}{\binom{100}{10}} + \frac{\binom{80}{10} \binom{20}{0}}{\binom{100}{10}}$$

$$\approx 0.3630$$

Problem #5

$X \sim \text{Unif}(0, 30)$, $X = \#$ of minutes waited until bus arrives

(a)

$$P(X > 10) = 1 - P(X \leq 10) = 1 - \frac{10}{30} = \frac{2}{3}$$

(b)

$$P(X > 25 | X > 15) = \frac{P(X > 25)}{P(X > 15)} = \frac{\frac{5}{30}}{\frac{15}{30}} = \frac{1}{3}$$

Problem #6

(a) for component i , it has lifetime of $X_i \sim \text{Exp}(\lambda_i)$

for components $i = 1, 2, \dots, 10$ they have mean $\mu_i = 1.1, 1.2, \dots, 2.0$

$\lambda_i = \frac{1}{\mu_i}$ Therefore, There is lifetime of system Y

$$Y = \min\{X_1, X_2, X_3, \dots, X_{10}\} \sim \text{Exp}\{\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_{10}\}$$

(b)

$$\lambda_Y = \lambda_1 + \lambda_2 + \dots + \lambda_{10} \approx 6.687714$$

$$E[Y] = \frac{1}{\lambda_Y} = 0.149528 \text{ hours}$$

$$SD(Y) = \sqrt{\text{Var}(Y)} = \sqrt{\frac{1}{\lambda_Y^2}} = \frac{1}{\lambda_Y} = 0.149528 \text{ hours}$$

Problem #7

For Romeo, he has being late distribution of $X \sim \text{Exp}(\lambda)$

For Juliet, she has being late distribution of $Y \sim \text{Exp}(\lambda)$

$$\text{difference } D = X - Y \Rightarrow X = D + Y$$

$$f(d) = \int_0^\infty f_X(d+y) f_Y(y) dy, \quad f_X(x) = \lambda e^{-\lambda x}, \quad f_Y(y) = \lambda e^{-\lambda y}$$

the integral is nonzero only when $y \geq 0$ and $d+y \geq 0$

for $d > 0$:

$$\begin{aligned} f_0(d) &= \int_0^\infty \lambda e^{-\lambda(d+y)} \cdot \lambda e^{-\lambda y} dy = \lambda^2 e^{-\lambda d} \int_0^\infty e^{-2\lambda y} dy \\ &= \frac{\lambda}{2} e^{-\lambda d} \end{aligned}$$

for $d < 0$:

$$y \geq -d$$

$$\begin{aligned} f_0(d) &= \int_{-d}^\infty \lambda e^{-\lambda(d+y)} \cdot \lambda e^{-\lambda y} dy = \lambda^2 e^{-\lambda d} \int_{-d}^\infty e^{-2\lambda y} dy \\ &= \frac{\lambda}{2} e^{\lambda d} \end{aligned}$$

$$\therefore f_0(d) = \frac{\lambda}{2} e^{-\lambda|d|} \quad d \in \mathbb{R}$$